

Vipul Shrivastav

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PROFILE SUMMARY

Software Developer and 2026 CS graduate who builds things that work in the real world — conveyor belt detection systems, CV pipelines, backend APIs. Currently at SmarDen Automation delivering production CV projects with 91.9% mAP on a custom YOLOv11 model. Strong in Python, DSA, and problem-solving. Open to software engineering roles across AI, CV, and backend.

EXPERIENCE

Software Developer Intern | SmarDen Automation, Haryana | June 2026 – Present

- Built a real-time product counting system for **Blinkit** using Python and OpenCV on a conveyor belt — handled dynamic lighting conditions affecting detection accuracy
- Developed carton box detection and counting system for **Maruti Suzuki** conveyor belt using Python, YOLOv11, and OpenCV
- Trained custom YOLOv11 model on self-annotated dataset — **91.9% mAP · 99% Precision · 88.5% Recall · 93.5% F1**
- Annotated 200+ images using Roboflow Label Assist; applied augmentation to expand dataset 4x for robust training
- Trained on Google Colab T4 GPU with Drive checkpointing to prevent progress loss across sessions
- Implementing active learning pipeline to scale annotated dataset from 200 to 1000+ images

EDUCATION

Bachelor of Technology – Computer Science & Engineering | 2022 – 2026

Maharana Pratap Group of Institutions, Kanpur

CGPA – 7.4/10

TECHNICAL SKILLS

Languages: Python, C, JavaScript

Computer Vision / ML: OpenCV · YOLOv11 · Object Detection · Roboflow · NumPy · Scikit-Image · Google Colab

Backend: FastAPI (learning) · Django · Flask · REST APIs · Node.js · Express.js

Frontend: HTML · CSS · JavaScript · React (basics)

Databases: MySQL · MongoDB ·

DSA: Arrays · Strings · Hashmaps · Two Pointers (active practice on LeetCode)

Tools: Git · GitHub · Postman · Roboflow · VS Code

Concepts: OOPs · DBMS · OS · SDLC

PROJECTS

Conveyor Belt Product Counting System — Blinkit | Python · OpenCV · MOG2 · NumPy

- Built real-time product counting system for Blinkit's industrial conveyor belt using Python and OpenCV — designed as a production-ready POC for Raspberry Pi deployment
- Implemented MOG2 background subtraction with morphological operations to detect and track moving products across a virtual counting line
- Solved real industrial problem — built lighting change detection using foreground pixel percentage to automatically pause and resume counting during sudden bulb ON/OFF events, eliminating false counts
- Optimized for real-time performance with ROI-based processing; supports USB camera, IP webcam, and video file inputs

- Modular codebase with clean separation — detector.py, counter.py — following production-grade code structure

Carton Box Detection System — Maruti Suzuki | Python · YOLOv11 · OpenCV · Roboflow · Google Colab

- Built modular Python-based carton box detection system for Maruti Suzuki conveyor belt using custom-trained YOLOv11 model and OpenCV
- Trained custom YOLOv11 model on self-annotated dataset — **91.9% mAP · 99% Precision · 88.5% Recall · 93.5% F1**
- Clean modular architecture — CartonBoxDetector class in detector.py — supports image, video, and live IP camera/webcam streams with zero code changes
- Full ML pipeline: image collection → Roboflow annotation → augmentation (4x expansion) → Colab T4 GPU training → inference
- Used active learning (Label Assist) to auto-annotate new images — reduced manual annotation effort by 80%
- Designed for future extension — conveyor belt tracking, object counting, and line crossing without modifying input stream scripts

Low-Light Image Enhancement System | Python · OpenCV · Scikit-Image · NumPy

- Developed an image enhancement system that improves low-light images using YUV color correction, CLAHE, and gamma adjustment
- Implemented optimized SSIM and PSNR image quality metrics using Scikit-Image
- Improved brightness, contrast, and visibility while preserving natural colors
- Designed a scalable pipeline capable of handling single images, batches, or real-time frames
- Wrote clean, modular, and reusable code following best practices
- Tested the system on multiple datasets and debugged issues to improve output accuracy
- Optimized image processing pipeline to improve efficiency and reduce noise

DSA & PROBLEM SOLVING

Active daily practice on LeetCode — covering Arrays, Strings, Hashmaps, Two Pointers, Sliding Window, Binary Search

- Problem Solving Certificate – HackerRank
- SQL 5-Star Badge – HackerRank

CERTIFICATIONS

- Problem Solving – HackerRank
- SQL (5-star Badge) – HackerRank
- MongoDB Basics – GeeksforGeeks